

OBLIG5

Kodeforståelse

Oppgave 1.1

Hva blir utskriften?

```
1 a = 4
2 b = 2
3 c = (a + b) / b
4
5 print(c ** 2)
```

$$C = (4+2)/2 = 3$$

$$3^2 = 9$$

Svar: 9

Oppgave 1.2

Hva blir utskriften?

```
1 numbers = []
2
3 for x in range(5):
4     numbers.append(x)
5
6 print(numbers)
7
8
9 numbers = []
10
11 for x in range(5):
12     numbers.insert(x, 0)
13
14 print(numbers)
```

Svar:

[0, 1, 2, 3, 4]

[0, 0, 0, 0, 0]

Oppgave 1.3

Hva blir utskriften?

```
1 class Course:
2     def __init__(self, name, number_of_students, study_points=10):
3         self.name = name
4         self.number_of_students = number_of_students
5         self.study_points = study_points
6
7     def get_description(self):
8         return f"The course {self.name} has {self.number_of_students} students" \
9             f" and {self.study_points} study points."
10
11
12 programming_1 = Course("Programming 1", 215)
13
14 print(programming_1.get_description())
```

Svar:

The course Programming 1 has 215 students and 10 study points.

Oppgave 1.4

Hva blir utskriften?

```
1 animals_in_zoo = {"honey badger": 2, "ape": 15, "zebra": 6, "giraffe": 4}
2
3 for animal in animals_in_zoo:
4     if animals_in_zoo[animal] < 5:
5         print(animal.title())
```

Svar:

honey badger

giraffe

Oppgave 1.5

Hva blir utskriften?

```
1 numbers = [5, 2, 3, 2, 4, 1]
2
3 sum_of_numbers = 0
4 for number in numbers:
5     if number <= 2:
6         sum_of_numbers += number
7     else:
8         sum_of_numbers += 1
9
10 print(sum_of_numbers)
```

Svar:

8

Oppgave 1.6

Hva blir utskriften?

```
1 animals = ["honey badger", "giraffe", "ape", "zebra"]
2
3 animals[1] = "elephant"
4
5 animals.sort()
6
7 animals = animals[:2]
8
9 for animal in animals:
10     print(animal)
```

Svar:

ape

elephant

Oppgave 1.7

Hva blir utskriften?

```
1 animals = ["Elephant", "Dog", "Cat", "Gorilla", "Dodo"]
2
3 animals = animals[1:3]
4
5 animals[0] = "Alligator"
6
7 animals.sort(reverse=True)
8
9 print(animals)
```

Svar:

Gorilla

Cat

Alligator

Oppgave 1.8

Hva blir
utskriften?

```
1 shopping_list = {}
2
3
4 def add_item(item_name, quantity=1):
5     if item_name in shopping_list.keys():
6         shopping_list[item_name] += quantity
7     else:
8         shopping_list[item_name] = quantity
9
10
11 add_item("Bread", 2)
12 add_item("Milk")
13 add_item("Milk", 2)
14 add_item("Bread", 2)
15 add_item("Eggs")
16 print(shopping_list)
```

Svar:

{"Bread": 4, "Milk": 3, "Eggs": 1}

Oppgave 1.9

Hva blir utskriften?

```
1 x = 0
2
3 for i in range(0, 5, 2):
4     x += i
5
6 print(x)
7
8 x = 0
9
10 for i in range(0, 5):
11     x += i
12
13 print(x)
```

Svar:

6

10

Oppgave 1.10

Hva blir utskriften?

```
1 a = 5
2 b = 2
3 c = 0
4
5 c += a ** b
6 print(c)
7
8 c += a % b
9 print(c)
10
11 c += a - b * 2
12 print(c)
13
14 c //= b
15 print(c)
```

Svar:

25

1

1

0

Oppgave 1.11

Hva blir utskriften?

```
1 class Game:
2     def __init__(self, name, genre, age_rating=18):
3         self.name = name
4         self.genre = genre
5         self.age_rating = age_rating
6
7     def description(self):
8         return f"The game {self.name} is of the genre {self.genre} and has an age rating of {self.age_rating}"
9
10
11 game1 = Game("Hades", "Rogue-lite", 12)
12 game2 = Game("God of War", "Action")
13 print(game1.age_rating)
14 print(game2.description())
```

Svar:

The game Hades is of the genre Rouge-lite and has an age rating of 12

The game God of War is of the genre Action and has an age rating of 18

Oppgave 1.12

Hva blir utskriften?

```
1 randomList = [1, 'a', 2, 'b', 3]
2 result = 0
3
4 for entry in randomList:
5     try:
6         result += int(entry)
7     except ValueError:
8         print("A ValueError occurred.")
9
10 print(f"The sum is: {result}")
```

Svar:

A ValueError occurred.

A ValueError occurred.

The sum is: 6